



A review on the recent progress on photodetectors

Xiaojuan Lian¹ , Limeng Luo¹, Minghao Dong¹, Ziyang Miao¹, Xi Qi¹, Zhikuan Cai^{1,*}, and Lei Wang^{1,*}

¹ The College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210023, China

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ABSTRACT

Photodetectors play a pivotal role in modern technology, particularly in applications requiring precise light detection and imaging. Traditional photodetectors, however, are often constrained by their fixed wavelength ranges, making them inadequate for multi-band detection needs. To address this limitation, phase change materials (PCMs) have emerged as a promising solution. PCMs are notable for their high sensitivity, rapid response times, and their capacity to detect a broad spectrum of wavelengths. These materials can transition between crystalline and amorphous states, allowing real-time adjustment of the detection wavelength. Furthermore, integrating surface plasmon resonance with photodetectors significantly enhances their performance by boosting light absorption and signal strength. This review delves into the fundamental principles, benefits, and challenges of various types of photodetectors. It specifically highlights traditional photodetectors, plasmon-enhanced photodetectors, and recent advancements in PCM-based photodetectors. By examining these innovations, we aim to provide valuable insights that could shape the future development and application of photodetector technology, ensuring they meet the evolving demands of various technological fields.

Introduction

In the modern field of technology, photodetectors are a crucial class of optoelectronic devices, playing a key role in converting light signals into electrical signals. These devices are not only essential in scientific research but also hold a critical position in various high-tech industries such as optical communication, medical imaging, environmental monitoring, defense

security, and solar cells [1–5]. However, traditional photodetectors face a series of challenges in meeting increasingly complex optoelectronic detection tasks. Firstly, traditional photodetectors are often limited by a fixed wavelength range, constraining their application in multi-band detection tasks. For instance, commonly used silicon photodetectors are primarily employed for detecting visible and near-infrared light [6, 7], while gallium nitride photodetectors exhibit

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Address correspondence to E-mail: whczk@njupt.edu.cn; leiwang1980@njupt.edu.cn

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excellent performance in the ultraviolet and blue light range [8–10], and indium gallium arsenide photodetectors demonstrate enhanced sensitivity in the near-infrared and short-wave infrared range [11–13]. In certain scenarios, simultaneous detection of light signals at different wavelengths is required, and traditional photodetectors necessitate the use of different devices to accomplish this task, thereby increasing system complexity. Secondly, the traditional integration approach of photodetectors often requires combining multiple narrow-band photodetectors to achieve multi-band detection. This method not only increases system complexity but may also lead to performance degradation and signal crosstalk issues. Particularly in the fields of communication, medical imaging, and defense, the demand for high-performance photodetectors in specific wavelength bands continues to grow, exacerbating these challenges. Therefore, the field of photodetectors urgently requires a novel solution to address the aforementioned challenges and demands. This solution should possess ultra-wide-band detection capability, enable wavelength tuning to adapt to light signals across different wavelength ranges. Additionally, the new photodetector should reduce system complexity, enhance performance, and meet the requirements of multi-band detection tasks.

Currently, photodetectors based on two-dimensional (2D) materials, such as graphene and transition metal dichalcogenides (TMDs), are receiving significant attention. These photodetectors exhibit a broader spectral response range, higher sensitivity, and faster response times, making them highly promising for multi-band detection and high-performance applications [14–18]. However, the spectral response tuning for different wavelengths in many 2D material-based photodetectors is primarily achieved through band engineering or chemical design, which results in complex manufacturing processes and significant optical losses [19–21]. Additionally, organic photodetectors (OPDs) are gaining attention for wearable sensing applications due to their potential for low-cost, large-area, and flexible electronics [22–24]. However, OPDs face several drawbacks, including relatively low carrier mobility, which limits response speed, and instability under environmental conditions such as humidity and oxygen, which can degrade performance over time [25, 26]. Furthermore, the efficiency and lifespan of OPDs still lag behind their inorganic counterparts, posing challenges for their widespread adoption in high-performance applications [27]. Against this

backdrop, researching and developing high-performance new photodetectors with tunable wavelengths has become an international research focus. These new photodetectors are expected to effectively address the evolving tasks of photodetection, advance modern technology fields, and provide more powerful tools for scientific research and high-tech industries.

In recent years, phase change materials (PCMs) have emerged in the field of optoelectronic detectors, demonstrating outstanding performance, including high sensitivity, fast response time, controllability, low energy consumption, long life, and a wide range of wavelength responses [28–34]. These characteristics make them highly anticipated detector materials. PCMs belong to a special type of non-volatile optoelectronic material, whose atomic structure can reversibly change between crystalline and amorphous states under the action of external electric pulses or light signals [28]. In many optical and optoelectronic applications, it is necessary to detect light signals of different wavelengths. Traditional optoelectronic detectors typically have a fixed spectral response range and cannot adapt to light signals of different wavelengths. In contrast, optoelectronic detectors based on PCMs have the ability to dynamically adjust the detection wavelength in real-time [31–34]. Additionally, the surface plasmon resonance (SPR) effect provides an additional opportunity to enhance the performance of optoelectronic detectors. By using surface plasmons, higher light absorption efficiency and signal enhancement can be achieved [32–37]. Improvements in different surface plasmon materials have strengthened the response speed of optoelectronic detectors and provided more choices for the design of new types of optoelectronic detectors. Researchers are constantly striving to improve the performance of surface plasmon optoelectronic detectors, especially in terms of material selection and device design. Figure 1 presents the main content of this review paper, including traditional photodetectors, SPR photodetectors, and PCMs-based photodetectors.

This review article aims to comprehensively present the current advancements in photodetector technology, including research progress, new material innovations, and the latest developments in related technologies. We will explore the operational principles, advantages, and limitations of various photodetector types, with a specific focus on traditional models, plasmon-enhanced designs, and the most recent progress in PCM-based photodetectors. By analyzing

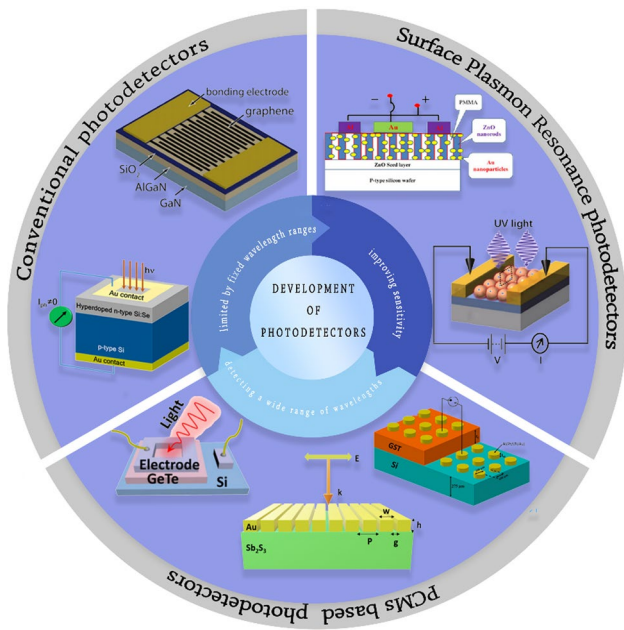


Figure 1 The primary focus of this review article encompasses conventional photodetectors, SPR photodetectors, and PCMs-based photodetectors.

the performance advantages of different photodetector types, we aim to provide references for the future

development of photodetectors. Furthermore, we will discuss potential research directions to enhance the performance and expand the application areas of photodetectors. Through this review, we seek to offer valuable insights into the future development of photodetector technology.

Conventional photodetectors

Conventional photodetectors play a vital role at the intersection of optics and electronics, and are widely used in a variety of fields. Operating on the fundamental principle of the photoelectric effect, these detectors possess the capability to convert incident photons into electrical signals, facilitating the detection and measurement of optical signals. The conventional photodetectors operate based on fundamental principles, including photoconductivity, grating pressure, and the photovoltaic effect [38]. Figure 2 provides a visual representation of their operational principles, accompanied by corresponding photoelectric characteristic curves. Specifically, Fig. 2a depicts the photoconductivity effect, with its associated photoelectric characteristic curve labeled as Fig. 2d. The grating pressure effect is illustrated in Fig. 2b, complemented by its

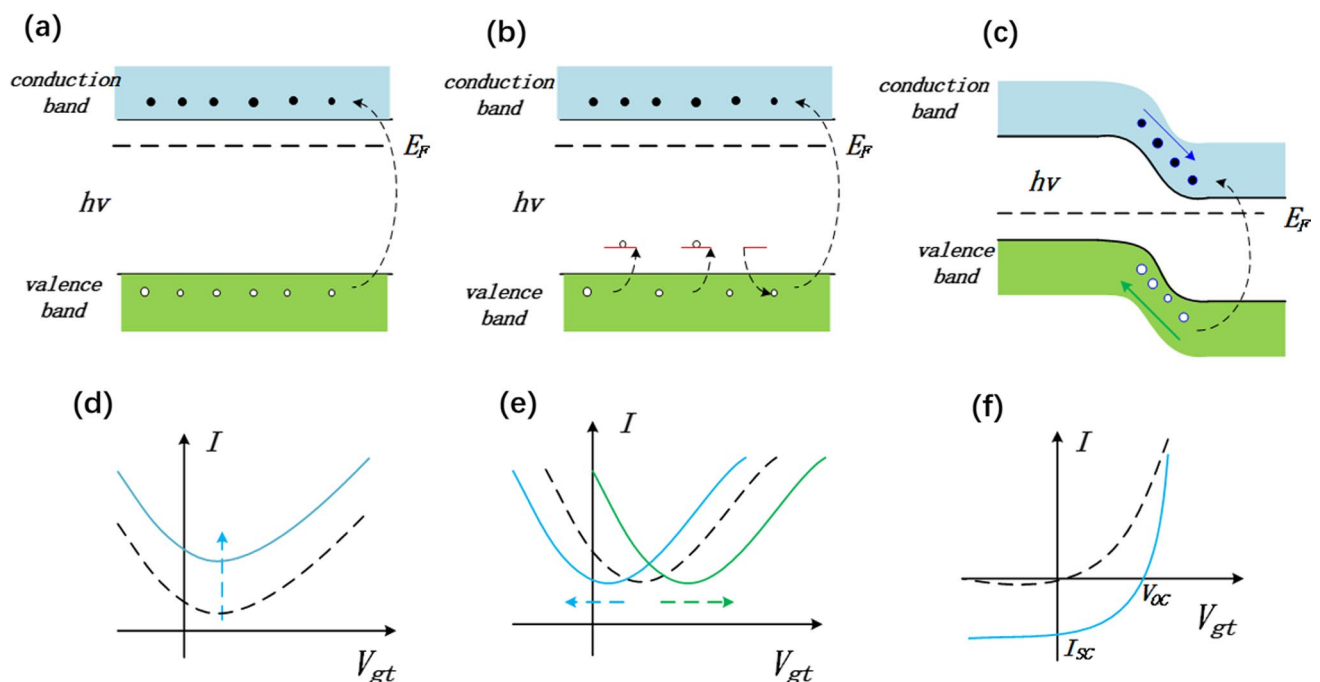


Figure 2 Conventional photodetectors operate based on several fundamental principles, including. **a** The photoconductivity effect, **b** the grating pressure effect, **c** the photovoltaic effect, **d–f**

illustrate the photoelectric characteristic curves corresponding to the photoconductive effect, grating pressure effect, and photovoltaic effect, respectively.

photoelectric characteristic curve denoted as Fig. 2e. Lastly, Fig. 2c showcases the photovoltaic effect, with its corresponding photoelectric characteristic curve represented as Fig. 2f.

In 1839, French physicist Alexandre Becquerel observed the photoelectric effect, a phenomenon indicating the generation of electric current when light impacts a material's surface. This pivotal discovery marked a significant advancement in photoelectric detection technology [39]. Subsequent inventions, such as the photodiode and phototube [40], along with the introduction of semiconductor materials like silicon photodiodes and silicon phototransistors, have driven this technology forward. Today, predominant materials for conventional photodetectors mainly comprise inorganic semiconductors such as silicon (Si), gallium nitride (GaN), indium gallium arsenide (InGaAs), and two-dimensional (2D) materials like graphene and perovskite. The choice of materials depends on factors such as the application's wavelength range, performance criteria, and cost considerations.

Inorganic material photodetectors

Si photodetectors are among the most commonly used photodetectors, and their primary advantage lies in their exceptional response to visible and near-infrared light [41–44]. These photodetectors are relatively cost-effective to manufacture, offering high resolution and high speed, along with a high quantum efficiency. Jurgen Michel et al. [45] investigated the Germanium-on-Silicon (Ge-on-Si) photodetector technology, and the structural design of the Ge-on-Si photodetector is depicted in Fig. 3a. Their research primarily centered on the feasibility of integrating Ge-on-Si photodetectors with CMOS technology. This integration creates highly integrated electronic and optoelectronic devices, offering the potential for improved performance in high-speed optical communication systems. They also emphasized the importance of integrating Ge-on-Si photodetectors with silicon waveguide technology, allowing efficient optical signal input while reducing the photodetector's footprint for improved performance. This is particularly advantageous in high-density optical communication systems. Furthermore, Ge-on-Si photodetectors operate within the eye-safe wavelength band of 1500 nm, achieving close to 100% quantum efficiency in this range [45], as shown in Fig. 3b. This implies efficient conversion of input optical signals into electrical signals with minimal

energy loss, which distinguishes them from materials with lower quantum efficiency that may require higher optical power for equivalent.

Yonder Berencén et al. demonstrated room-temperature short-wavelength Infrared (SWIR) p-n photodiodes from single-crystal Si hyper doped with Se using a robust and reliable non-equilibrium processing consisting of ion implantation followed by millisecond-range flash lamp annealing (FLA). Its structure is shown in Fig. 3c [7]. This approach is repeatable, low-cost, allows for scalability and incorporates non-equilibrium Se concentrations as high as $9 \times 10^{20} \text{ cm}^{-3}$ in a way that the material's pristine structure is preserved. It has also the advantage of preventing surface segregation, dopant redistribution and clustering. The SWIR photoresponse was demonstrated to be related with known Se deep-energy levels in Si. The photoresponse of hyperdoped Si can be extended up to 3100 nm (0.4 eV) according to Fourier transform infrared (FTIR) absorbance spectra. Figure 3d shows the normalized photocurrent of the studied Se atom concentration photodetector under a reverse bias voltage of 1 V [7]. These photodetectors are also suited for simultaneously covering both visible and SWIR spectral range. We believe that this technology can start to be competitive once the conservative barrier of the 1% in quantum efficiency can be overcome. However, Si-based photodetector devices exhibit certain drawbacks, including inadequate photon energy for telecommunication band light detection and the generation of substantial dark currents. To address these issues, doped Schottky photodetectors have been introduced, but they suffer from dark currents-related limitations [46].

GaN photodetectors are superior in the ultraviolet (UV) and blue light ranges, offering high electron mobility and thermal stability. This makes them ideal for applications like UV LED, laser monitoring, and UV spectral analysis [10]. Figure 3e schematically shows different kinds of the photodetectors realized using the GaN materials system [47]. Shur et al. [47] reported on AlGaIn photoconductive detectors, which consisted of a 0.5- μm -thick AlGaIn layer deposited over a basal plane sapphire substrate (with a usual buffer layer). Figure 3f illustrates the spectral responsivity of GaN-related photodetectors, particularly $\text{Al}_x\text{Ga}_{1-x}\text{N}$ [47], with lines a, b, c, and d corresponding to x values of 0.61, 0.55, 0.46, and 0.05, respectively.

Bhishma Pandit et al. [48] explored a bifunctional UV photodetector with a substantial UV–visible

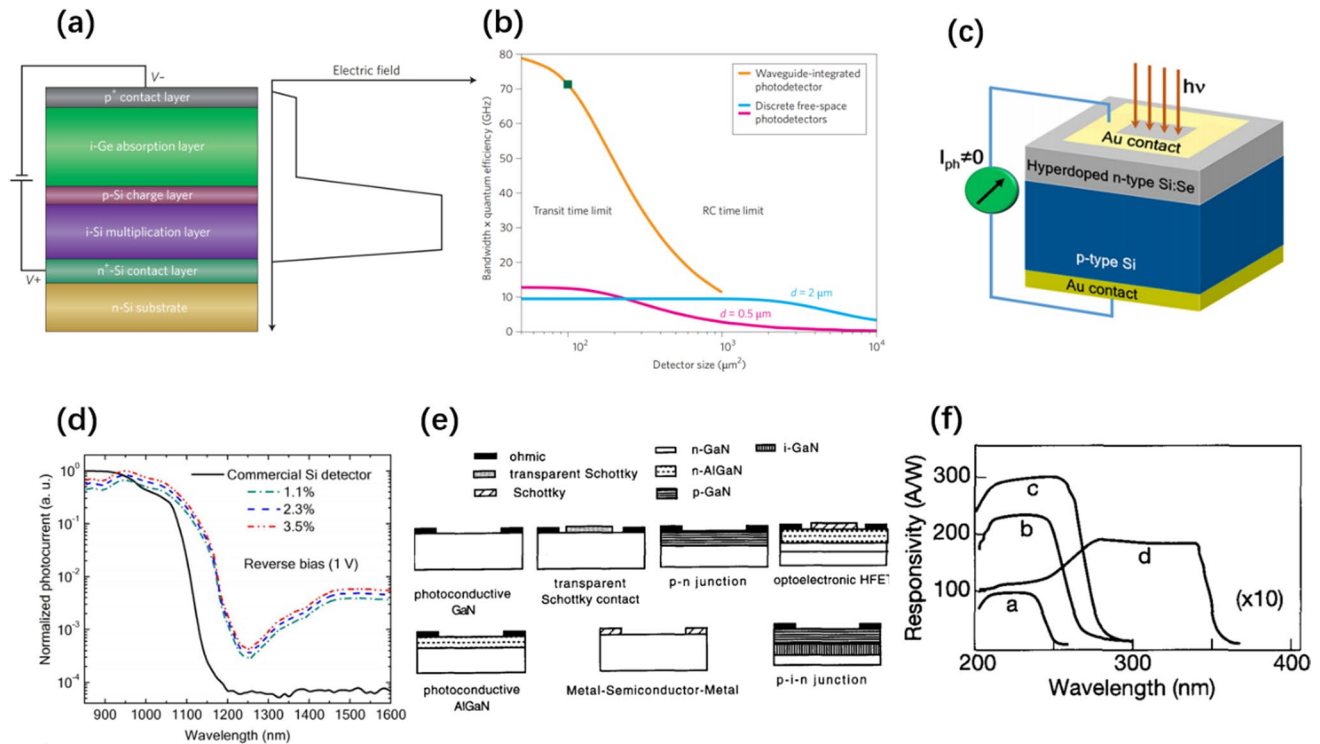


Figure 3 **a** Cross-sectional view of a Ge/Si avalanche photodiode (APD) with a separate-absorption-charge-multiplication (SACM) structure, highlighting the internal electric field distribution [45]. **b** Relationship between bandwidth-quantum efficiency product and detector area for various Ge detector designs at a wavelength of 1550 nm [45]. **c** Cross-section diagram of Se-hyperdoped Si p-n photodiode devices [7]. **d** Normalized photo-

current of photodetectors with different atomic Se concentrations at a reverse bias of 1 V, with the spectral response of a commercial Si photodiode shown for reference. All spectral photo responses are normalized to the peak Si photodiode response [7]. **e** Illustration of GaN-based photodetectors [47]. **f** Spectral responsivity of Al_xGa_{1-x}N photoconductors at various alloy compositions [47].

rejection ratio. They incorporated 2D graphene electrodes into an AlGaIn/GaN heterostructure, as shown in Fig. 4a. This integration facilitates two distinct photocurrent generation mechanisms: photovoltaic and photoconductive. The photodetector exhibits both modes within a single device, allowing selectable via bias voltage to cater to various UV light detection requirements. Nonetheless, GaN photodetectors have notable limitations. Their energy band structure leads to carriers with a higher effective mass and poorer transport properties, resulting in low electric field mobility and poor high-frequency performance.

Xuanzhang Li et al. [49] demonstrate high performance visible-SWIR flexible photodetector based on large-area InGaAs/InP PIN structure. The structure of this photodetectors is shown in Fig. 4b. Its operating wavelength from 640 to 1700 nm with a high detectivity of 5.18×10^{11} cm Hz^{1/2}/W and fast response speed at a wavelength of 1550 nm by using a simply top-to-down fabrication process. The optoelectrical

performances are stable as the photodiodes are exposed to bending cycles with a radius of 15 mm up to 1000 times. Figure 4c shows the spectral response ranges of the InGaAs-based membrane photodiode (red line) and the InGaAs PIN rigid photodiode (black line, as a counterpart) with the same area fabricated from the same epitaxy wafer [49]. However, despite these advancements, the development and application of InGaAs photodetectors encounter challenges, including a limited detection range and elevated dark current [13].

2D material photodetectors

Traditional semiconductors like Si and InGaAs photodetectors face limitations in modern electronics and photonics, including spectral coverage, low resolution, lack of transparency and flexibility. In contrast, emerging 2D materials such as graphene and TMDs, and their hybrid systems offer solutions to these

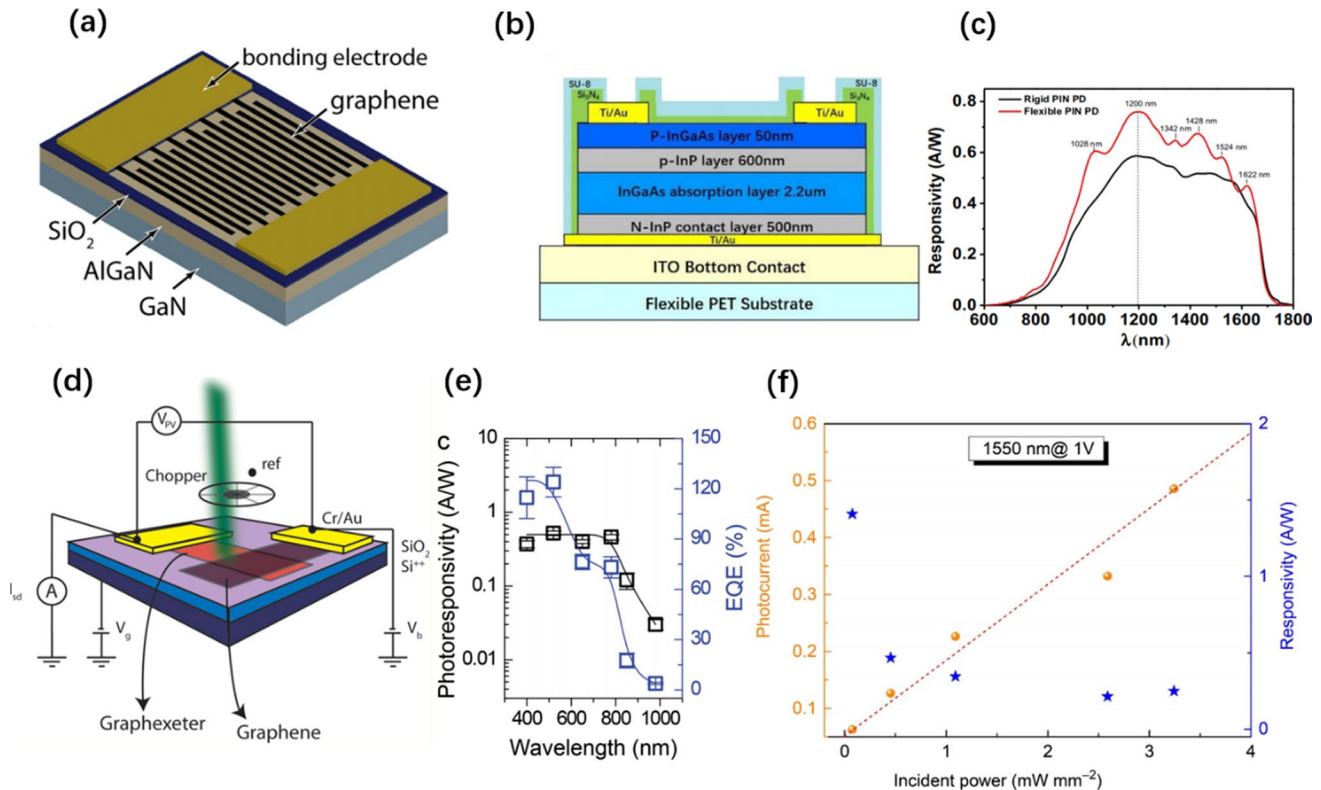


Figure. 4 **a** Schematic of the Gr/AlGaIn/GaN photodetector structure [48]. **b** Structure of a high- performance visible-SWIR flexible photodetector based on a large-area InGaAs/InP PIN structure [49]. **c** Spectral response ranges of the InGaAs-based membrane PD (Red line) and the InGaAs PIN rigid PD (Black line, as a counterpart) with the same area fabricated from the

same epitaxy wafer [49]. **d** Schematic of the device design and the experimental setup for photovoltage spectroscopy [51]. **e** Responsivity of the heterojunction device under 1550 nm wavelength illumination [54]. **f** Graph depicting photoresponsivity and EQE concerning the wavelength of the illumination source [56].

challenges. These materials leverage mechanical flexibility, exceptional electronic and optical properties, and the ability for wafer-scale production and integration, addressing the shortcomings of conventional semiconductors [50].

Graphene has been extensively explored in the field of optoelectronic detection due to its distinctive zero bandgap energy band structure and ultra-high mobility. This material exhibits an ultra-wide spectral detection range, spanning from ultraviolet to terahertz, and an ultra-fast response speed with a working bandwidth up to 500 GHz [1]. Freddie Wither et al. have successfully developed photovoltaics in a novel graphene device, incorporating FeCl₃ intercalated few-layer graphene (FeCl₃-FLG) and pure graphene, as illustrated in Fig. 4d [51]. The intercalation of FeCl₃ enhanced graphene's electrical conductivity, resulting in a substantial carrier concentration of approximately $\sim 9 \times 10^{14} \text{ cm}^{-2}$. In

another notable accomplishment, Fengnian Xia et al. demonstrate ultrafast transistor-based photodetectors made from single-layer and few-layer graphene [52]. The photoresponse does not degrade for optical intensity modulations up to 40 GHz, and further analysis suggests that the intrinsic bandwidth may exceed 500 GHz. Despite these advancements, it is crucial to acknowledge that graphene has its inherent limitations, including weak light absorption, rapid carrier recombination, and an inadequate gain mechanism, which can impact the overall device performance [53]. Youngbin Lee et al. introduced an inventive hybrid photodetector merging graphene with a CH₃NH₃PbI₃ perovskite layer [54]. This device demonstrates an extensive photoresponsivity range from 800 to 400 nm, accomplished through improved light absorption and the integration of light gating mechanisms within the hybrid setup. Figure 4e depicts the photoresponsivity (black line) and external quantum

efficiency (EQE, blue line) varying with incident light wavelength [54].

On the other hand, perovskite materials have garnered significant attention for their good photore-sponse performance due to their high carrier mobility, long diffusion length, and high stability [55]. Wei Hu et al. [56] have successfully engineered a perovskite/germanium heterojunction with a well-optimized interface. This innovative design effectively addresses the individual limitations of perovskite and germanium materials, harnessing their respective strengths to achieve wide-band photodetection. The impressive device responsivity is visually presented in Fig. 4f. Jichao Yu et al. [57] report a self-powered photodetector based on a ZnO/CH₃NH₃PbI₃ heterojunction and a MoO₃ hole-transport layer. The organolead iodide perovskite photodetector is sensitive to broadband wavelengths from the ultraviolet light to the entire visible light region (250–800 nm), showing a high photo-responsivity of 24.3 A W⁻¹ and a high detectivity value of 3.56×10^{14} cm Hz^{1/2}/W at 500 nm without external bias voltage. Meanwhile, we found that the photodetective performances are closely related to the thickness of the MoO₃ layer, which acts as a hole-transport layer and an electron blocking layer and can effectively decrease the recombination of holes and electrons. Additionally, the as-fabricated photodetector exhibits good stability and only 9.3% photoelectric response current decay after a 3-month illumination test. The high detectivity and responsivity of such a ZnO nanorod/perovskite heterojunction are clearly demonstrated and should be widely applicable to other photoelectric detection devices.

In addition, MXene, an emerging 2D material, was first discovered by Drexel University in 2011 [58], demonstrating excellent performance in the field of photodetectors. MXene usually exists in the form of M_{n+1}X_nT_x, where M represents early transition metals (such as Ti, V, Cr, Nb, Mo, etc.), X represents carbon or nitrogen, and T_x represents functional groups (mainly –O, –F, –OH) [59]. Compared with other 2D materials, MXene has good hydrophilicity and can complete the preparation process of MXene films through simple, multifunctional, and low-cost solution processing techniques. Recently, significant progress has been made in the field of MXene photodetectors. For instance, Kiana Montazeri et al. [18] successfully prepared ultra-high-speed, high-sensitivity MXene/semiconductor/MXene structured photodetectors (Fig. 5a, b). Figure 5c, d show the current–voltage

(*I*–*V*) curves in the dark and under illumination from both continuous wave (CW) and pulsed lasers at 800-nm wavelength [18]. The dark current of 50 fA/μm² is quite low, reflecting the unique properties of the MXene-semiconductor. The photocurrent under various optical powers and biases is shown in Fig. 5e, f for CW and pulsed cases [18], respectively. These detectors achieved a record responsivity of up to 1.04 A W⁻¹ at low light power, an internal quantum efficiency greater than 230%, a dynamic range greater than 105.6 dB, and a response time in the range of 25–150 ps. Dhinesh Babu Velusamy et al. [60] developed photodetectors using five different MXenes, with molybdenum carbide MXene (Mo₂CT_x) showing the best performance. Mo₂CT_x MXene films deposited on paper substrates exhibit broad photoresponse in the 400–800 nm range, with high sensitivity (up to 9 A W⁻¹), high detectivity (about 5×10^{11} Jones), and reliable photoswitching characteristics at a wavelength of 660 nm. Chuqiao Hu et al. [61] designed a Ti₃C₂T_x/RAN near-infrared photodetector based on organic–inorganic van der Waals-hydrogen bond heterostructure. The Ti₃C₂T_x MXene film electrode and the organic RAN film material have good contact, and the prepared detector exhibits good mechanical stability. After 5000 bending cycles, there is no significant change in photocurrent and dark current. Weidong Song et al. [62] reported a Ti₃C₂ MXene/Si heterojunction photodetector with ultra-high specific detectivity (2.03×10^{13} Jones) and significant responsivity (402 mA W⁻¹), which does not diminish with increasing light power at zero external bias. This characteristic is achieved through a chemically regenerated SiO_x interfacial layer and controlled Ti₃C₂ MXene thickness. The photodetector exhibits a high light-switching ratio of more than 10⁶, a peak external quantum efficiency of 60.3%, and maintains an extremely low dark current at 0 V bias.

Despite the widespread utilization and continuous progress of graphene and perovskite materials in photodetectors, both exhibit noteworthy limitations. Graphene's inherent lack of a bandgap, coupled with cost and technical complexities in its production, presents challenges for precise detection within specific wavelength ranges [38]. Additionally, there is still a significant gap in device performance between low-dimensional lead-free perovskite and 3D lead based perovskite, which further hinders the ultimate optical and electrical performance and its optoelectronic applications [55].

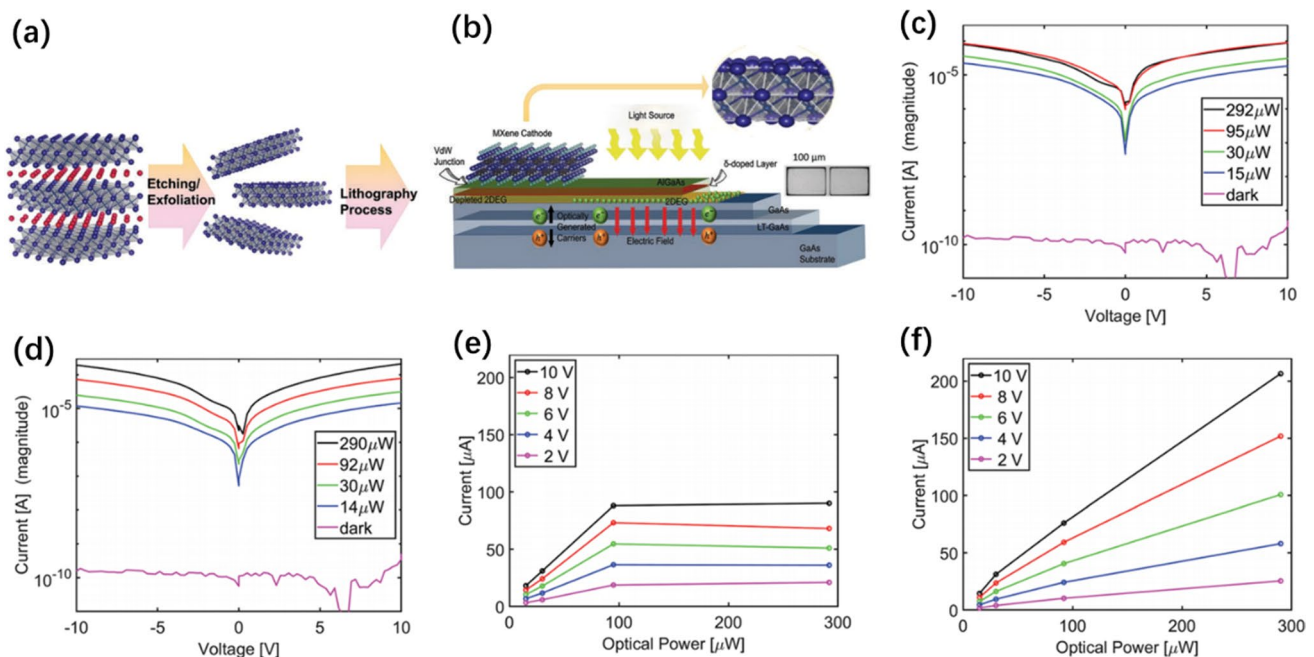


Figure 5 **a** Shows the synthesis process of single/few-layered MXene flakes by the selective etching of the A-layers from Ti_3AlC_2 [18]. **b** Depicts the structure of the fabricated device as detailed above [18]. **c** Presents the I–V curves under a CW laser [18]. **d** Illustrates the I–V curves under a pulsed laser with a 0.9 ps duration and 76 MHz repetition rate [18]. **e** Shows the

photocurrent versus optical power characteristics of the BE/MX-S-MX photodetector at different bias voltages using a CW laser [18]. **f** Displays the photocurrent versus optical power characteristics of the BE/MX-S-MX photodetector at different bias voltages using a pulsed laser [18].

Table 1 provides a comprehensive overview of the strengths and weaknesses inherent in these conventional photodetectors.

SPR photodetectors

SPR is an optical phenomenon that describes the oscillation of electrons on a metal surface when exposed to light [63]. It acts as a quasi-particle characterizing the collective movement of plasmons on

the metal surface. Plasmons can be primarily categorized into propagating surface plasmons (Fig. 6a) and local surface plasmons (Fig. 6b) based on their propagation properties [64]. This resonant oscillation leads to the interaction of photons with specific frequencies with the metal surface. When light of a particular frequency illuminates the metal surface, the free electrons on the metal surface interact with the incident photons, resulting in either resonance-driven absorption or scattering of the incident photons. Furthermore, SPR can locally

Table 1 Performance comparisons of different kinds of traditional photodetectors

Materials	Responsivity (A W^{-1})	Response time (ms)	Dark current (μA)	Detectivity (Jones)	EQE (%)	References
Ge	0.75	—	0.14	—	—	[45]
Se-hyperdoped Si	7.2×10^{-5}	—	57.3	—	6×10^{-3}	[7]
GaN-NT/Si	2.47	10^{-6}	0.12	1.95×10^{11}	—	[10]
GaN	12	0.67	—	—	—	[47]
InGaAs/InP	0.53	—	—	5.18×10^{11}	42.46	[49]
Perovskite/graphene	180	87	—	10^9	—	[54]

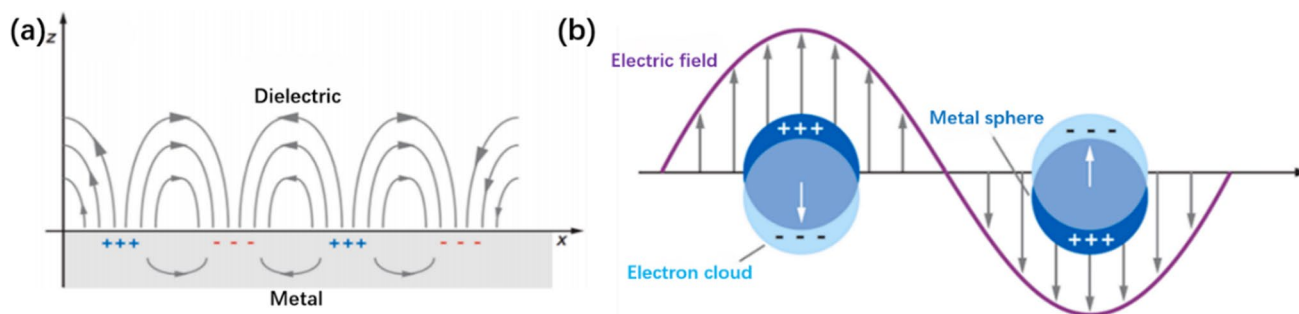


Figure 6 Basic principles of surface plasmons [64]. **a** Schematic diagram of a propagating surface plasmon. **b** Schematic diagram of a localized surface plasmon.

enhance electromagnetic field energy, concentrating the light's energy within subwavelength scales near the metal surface. This forms the foundation of surface-enhanced spectroscopy techniques and is a significant subject in optical research.

SPR offers unique advantages, including field enhancement, tunable resonance wavelengths, light energy convergence, and overcoming the traditional limitations of optical diffraction [65, 66]. It serves as a potent tool for interactions between light and matter, significantly enhancing light absorption. Traditional photodetectors improved with various surface plasmon materials exhibit increased responsivity to varying degrees. Notably, traditional photodetectors currently enhanced by SPR mainly include oxide-based photodetectors, nitride-based photodetectors, and 2D material-based photodetectors. In light of this categorization, we elaborate on photodetectors

utilizing surface plasmon materials for three distinct materials: zinc oxide (ZnO), graphene, and GaN (Tables 2, 3).

ZnO SPR photodetectors

ZnO is a wide-bandgap semiconductor known for its advantages, including high saturation drift velocity, excellent radiation resistance, and a high breakdown voltage. These properties make ZnO a popular choice in the manufacturing of UV photodetectors [67–70]. However, ZnO UV photodetectors have faced a persistent challenge due to their low responsivity, limiting their practical applications [71, 72]. To address this limitation, researchers have explored the use of SPR nanostructures.

SPR nanostructures resonate at specific frequencies and possess subwavelength-scale features. They act

Table 2 Remarkable achievements of surface plasmon enhanced photoelectric detectors

Detection wave-length	Materials	Responsivity	EQE	Detectivity (Jones)	Photoresponse enhancement	Surface plasmon	References
340 nm	ZnO	0.16 A W^{-1}	—	—	2 times	Au NPs	[74]
365 nm	ZnO	—	—	1.45×10^{10}	—	Ag NPs	[75]
325 nm	ZnO	0.485 mA W^{-1}	—	27.49×10^{10}	3.25 times	Au NPs	[76]
385 nm	GaN	112 A W^{-1}	$3.6 \times 10^4\%$	2.4×10^{12}	—	Ag, Au NPs	[90]
355/615 nm	GaN	—	—	—	54/64 times	Au NPs	[92]
365 nm	GaN	0.82 A W^{-1}	—	—	10 times	MoS_2	[93]
355 nm	GaN	670 A W^{-1}	—	1.48×10^{15}	—	Al NPs	[91]
382 nm	TiO_2/GaN	44.6 A W^{-1}	$8.65 \times 10^3\%$	1.57×10^{14}	—	Pt NP	[94]
405/450/532 nm	Graphene	$0.12/0.34/0.24 \text{ A W}^{-1}$	—	—	1.4/2.7/2.1 times	AgAu NPs	[82]
514 nm	Graphene	2.2 A W^{-1}	$\sim 1.5\%$	—	4 times	Au NPs	[84]
375 nm/4 μm	Graphene	$10^9/45 \text{ A W}^{-1}$	—	—	—	Si QDs	[86]
2.75 μm	Graphene	0.13 A W^{-1}	—	—	—	Si	[85]

Table 3 The performance comparisons of different phase change materials-based photodetectors

Materials	E _g (eV)	Wave-length (nm)	Intensity (mW/cm)	Responsivity (A W ⁻¹)	rise/fall time	detectivity (Jones)	I _{Photo} current/I _{Dark} current	References
A-GST	0.65	1550	1	~48	—	—	> 100	[37]
C-GST	0.5	1150	1	~63	—	—	—	[109]
A-GST/C-GST	0.65/0.5	632	8	2.3	~0.3 us/~0.15 us	—	—	[111]
A-Sb ₂ S ₃ /C-Sb ₂ S ₃	1.5/2.3	532	0.03	~65	—	2.1 × 10 ¹⁴	< 2.2 × 10 ⁴	[123]
A-Sb ₂ S ₃ /C-Sb ₂ S ₃	1.7–1.8/1.7–2.8	450	10–60	~112	—	—	—	[124]
A-GeTe/C-GeTe	0.85/0.84	514	—	~100	—	1–8 × 10 ¹¹	—	[127]

as subwavelength scattering sources and nanoantennae, enhancing optical detection and responsiveness, thereby increasing the sensitivity of photodetectors. Different surface plasmons have been applied to enhance the responsivity of ZnO UV photodetectors to varying degrees. Some researchers have successfully employed SPR with metallic nanoparticles (NPs) such as Al, Au, and Ag to achieve resonant enhancement of ZnO UV photodetectors' responsivity [73–76]. In

Fig. 7a, b, you can see the schematic structure of ZnO photodetectors covered with metallic nanoparticles. Due to the presence of metal nanoparticles, when light at specific frequencies illuminates ZnO nanorods (ZnO NRs), the free electrons on the surface of the metal nanoparticles interact with the light field, resulting in the excitation of surface plasmon resonance. This resonance effect amplifies the electromagnetic field in the vicinity of the metal nanoparticles, thus prompting

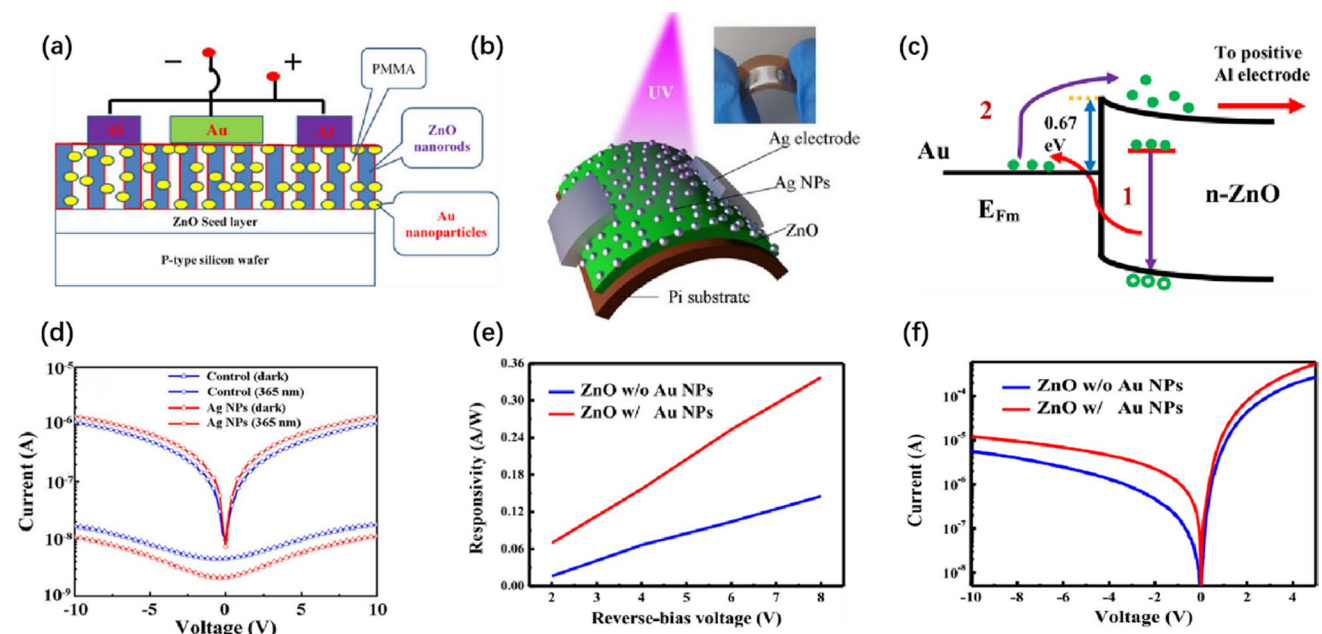


Figure 7 **a** Schematic structure of ZnO-based Schottky-barrier photodetector with Au NPs covering [74]. **b** Schematic structure of an Ag NP-modified ZnO UV photodetector [75]. **c** Band-diagram of Au NPs/n-ZnO NRs [74]. **d** *I*–*V* characteristics under dark conditions and 365 nm UV at 715 mW [75]. **e** Responsivity

versus reverse-bias voltage for the SB-PDs without and with Au NPs coverings, with an illumination wavelength of 340 nm [74]. **f** *I*–*V* characteristics of the fabricated SB-PDs without and with Au NPs coverings [74].

a substantial number of electrons to surmount the energy band barrier at the interface between the metal nanoparticles and ZnO. Upon reaching ZnO, these electrons can generate electron–hole pairs, thereby augmenting light absorption and enhancing the charge separation efficiency of the photodetector. As shown in Fig. 7c, a band diagram of Au NPs/n-ZnO NRs to clarify the improvement in UV responsiveness [74]. Additionally, Fig. 7d–f illustrate the characteristic curves of ZnO photodetectors with and without the coverage of metallic nanoparticles. It can be seen that the sample covering the metal nanoparticles had lower dark current and higher photocurrent than the control sample, which indicates that the sample covering the metal nanoparticles performed better than the control sample. The role of SPR has been substantiated, as ZnO photodetectors coated with metal nanoparticles demonstrate a considerably greater increase in responsivity under reverse bias voltage compared to those lacking the nanoparticle coating. This approach holds promise for improving the performance of ZnO UV photodetectors and expanding their practical utility.

Graphene SPR photodetectors

Graphene, a 2D material, exhibits wide optical absorption, ultra-high charge carrier mobility, and a tunable Fermi level, making it a promising prospect for applications in optical information transmission and photodetectors [77–80]. Graphene photodetectors offer rapid photoresponse at room temperature, covering an extensive bandwidth from 300 nm to 6 μm , encompassing UV, visible light, and the infrared spectrum [81–86]. Despite these advantages, graphene photodetectors face challenges such as low EQE and limited spectral selectivity.

Hao Suo et al. [82] developed a photodetector utilizing a metal-graphene-metal structure (Fig. 8a). Au and Ag nanoparticles, produced through annealing, induce localized SPR when exposed to light. This enhances photoresponse within the 400–470 nm and 510–620 nm wavelengths, resulting in a 2.8-fold improvement in responsivity. These enhancements lead to a 2 to 4-fold increase in responsivity in the specified wavelength ranges. Xiaomu Wang et al. incorporated graphene into silicon waveguides on Silicon-on-Insulator (SOI) to create a graphene/silicon heterostructure photodetector with a working range extending from the near-infrared to the mid-infrared [83], as depicted in Fig. 8b, c. Exploiting high refractive index contrast structures,

the interplay between graphene and CMOS-compatible silicon waveguides resulted in a photodetector with high responsivity. This waveguide effectively captures evanescent light propagating parallel to the graphene sheet, a crucial factor for enhancing detector performance and responsivity. Under conditions of room temperature and a 1.5 V bias, the photodetector achieved a substantial photoresponse of 0.13 A W^{-1} to 2.75 μm light, with a noted dependency of photocurrent on bias polarity. This phenomenon is ascribed to two distinct light absorption mechanisms: direct and indirect transitions of graphene at 1.55 μm and 2.75 μm . The research findings suggest that graphene's single-layer structure can directly absorb specific-wavelength light, and when combined with materials like silicon, can also employ indirect transitions to absorb light, thus broadening graphene's applications in detecting mid-infrared light.

Additionally, Yuan Liu et al. [84] introduced a graphene-gold nanoparticle detector, as shown in Fig. 8d. They conducted a comparative study of the spectra of graphene SPR with and without nanoparticle excitation, depicted in Fig. 8e, f. The photodetector containing gold nanoparticles achieved a photoresponse of up to 2.2 mA W^{-1} , while the one without gold nanoparticles exhibited a photoresponse of 0.42 mA W^{-1} . On average, the photoresponsivity increased by 400%. This research sheds light on the synergistic impact of graphene and gold nanoparticles in enhancing spectral performance and detector efficiency, offering novel insights for the advancement of photodetectors in the field.

GaN SPR photodetectors

GaN features a wide bandgap energy structure and exceptional optoelectronic properties, such as high carrier mobility. This makes GaN a viable material for optoelectronic applications spanning a range of wavelengths, including UV, visible light, and blue light [87–89]. However, certain GaN photodetectors exhibit relatively low EQE, signifying that only a small portion of incident photons undergo conversion into electrons, thereby reducing the sensitivity of the detectors.

Sundar Kunwar et al. [90] introduced a UV photodetector on GaN by integrating plasmonic nanoparticles through localized SPR, as depicted in Fig. 9a. The photoresponse characteristics of the photodetector are depicted in Fig. 9b–d. Remarkably, under a low bias of 0.1 V, it exhibited a photoresponsivity

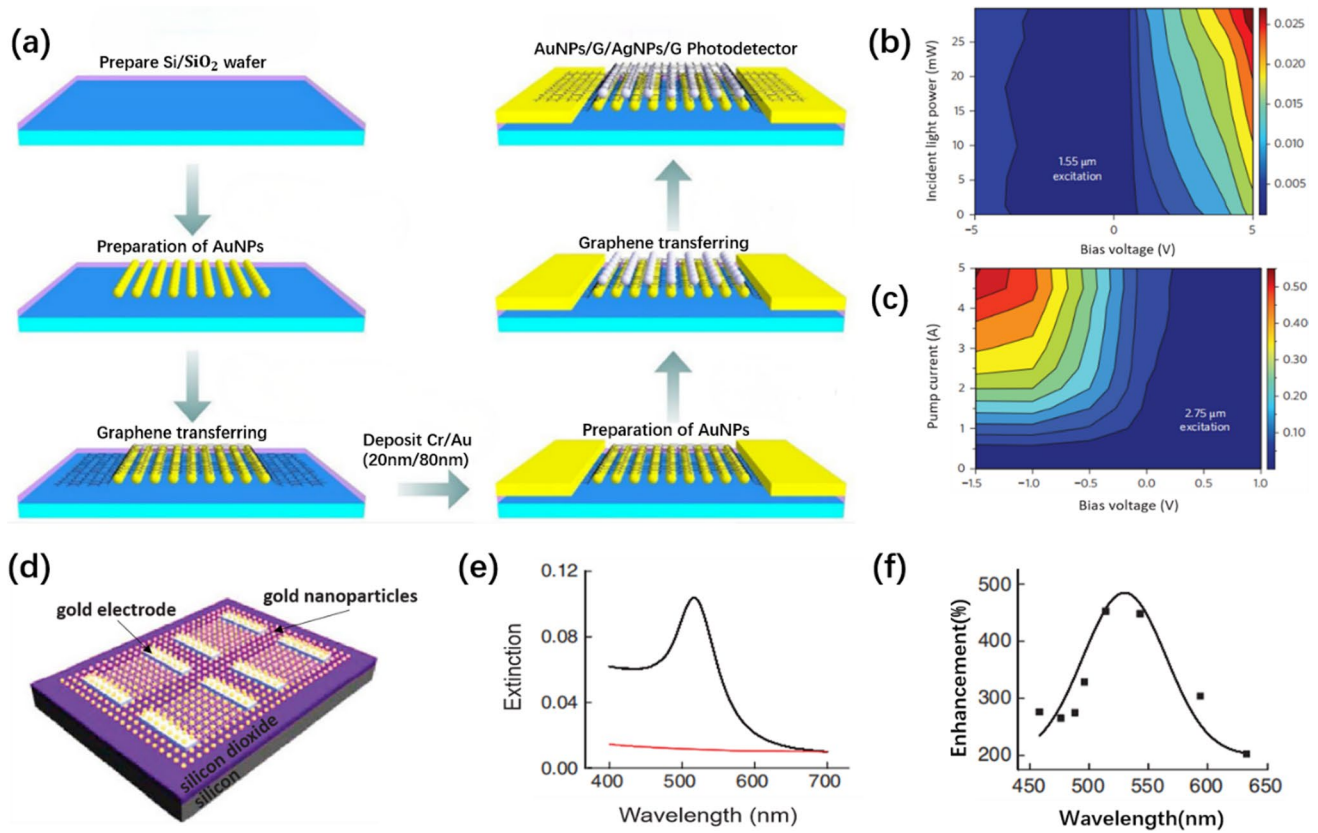


Figure 8 **a** Schematic illustration of the fabrication process for integrating plasmonic nanoparticles with graphene [82]. **b** Current as a function of incident-light power and electric bias at room temperature for a near-infrared photodetector (waveguide width: 500 nm) [83]. **c** Current as a function of pump current of incident light and electric bias at room temperature for a mid-

infrared photodetector (waveguide width: 1 mm) [83]. **d** Schematic diagram of the graphene-based nanoparticle photodetectors [84]. **e** A map of equisocion excitation without nanostructures and with nanostructures [84]. **f** Photocurrent enhancement as a function of laser wavelength [84].

of 112 A W^{-1} , a detectivity of 2.4×10^{12} Jones, and an EQE of $3.6 \times 10^4\%$. The AgAu alloy NPs detector also displayed rapid photoresponse with rise and fall times of less than 160 ms and 630 ms, respectively. This significantly improved the performance of the UV photodetector. Furthermore, Abhishek Dubey et al. [91] fabricated epitaxial single-crystal Al films on wide-bandgap semiconducting GaN, which serves as a high plasma frequency and low intrinsic loss metal for UV photodetectors. By designing periodic nano-hole arrays, the research team achieved localized SPR and excellent transmission. These plasmonic applications open up new possibilities for developing high-performance UV photodetectors based on GaN.

Ran Jia et al. [92] has introduced a dual-band plasmonic photodetector using GaN for highly sensitive detection of UV and green light. Their study confirmed that the improved photoresponse stems from

optically induced high-yield injection of hot electrons, as depicted in Fig. 9e. By incorporating Au nanoparticles and effectively harnessing the spontaneous polarization field inherent in GaN, the researchers significantly enhanced the detector's performance, as illustrated in Fig. 9f. This research holds significance for introducing a novel approach to designing and fabricating photodetectors, particularly when dealing with polar semiconductors and microstructure defects.

The photodetectors based on PCMs

Photodetectors are essential components in various applications, from data storage to optical communication. Traditional photodetectors, however, have faced challenges related to high power consumption, low efficiency, slow response times, and space

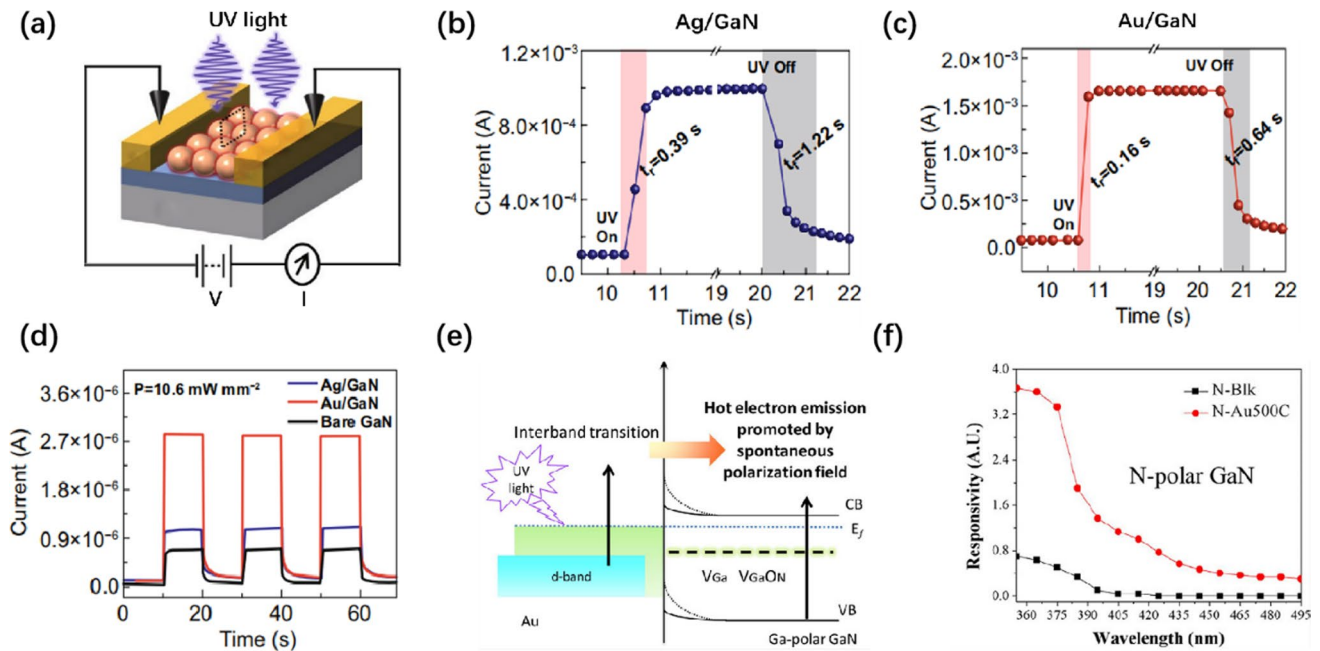


Figure 9 **a** Schematic of plasmonic metallic NPs based UV photodetector on GaN [90]. **b–c** Rise (t_r) and fall (t_f) time of the Ag and Au detectors upon switching on/off [90]. **d** Current–time (I – t) characteristic at 0 bias under 10.36 mW mm^{-2} UV illumination

[90]. **e** Schematic illustration of the interband transition with hot electrons injection for enhanced photoelectric responsivity to UV light [92]. **f** Photoelectric response curves in UV region for two N-polar samples [92].

requirements. To overcome these limitations, a novel class of materials has emerged, known as Phase Change Materials (PCMs), which have shown promise in revolutionizing the field of photodetection [33].

In the 1960s, scientists made the pioneering discovery of the phase transition properties of Ge–Sb–Te (GST), a well-known PCM. They observed that GST could shift between a crystalline state with high resistance and an amorphous state with low resistance [95]. As illustrated in Fig. 10a, PCMs undergo a SET process when stimulated by a small-amplitude, long-duration electric pulse, transitioning from the amorphous to the crystalline state. In contrast, a large-amplitude, short-duration electric pulse raises the temperature of the PCMs above their melting temperature (T_{melt}) due to Joule heating. This disrupts the chemical bonds in the crystalline phase. Rapid quenching then cools the molten PCM, preventing the re-bonding of atoms and leading to a transition back to the amorphous state, known as the RESET process [96]. Typical PCMs are composed of germanium (Ge), antimony (Sb), and tellurium (Te). Most current PCMs, represented in the Ge–Sb–Te ternary diagram (Fig. 10b), demonstrate fast crystallization speeds and significant changes in optical reflectivity [97]. These characteristics have laid

the foundation for exploring the potential of PCMs in photodetectors. During the 1980s, researchers began applying GST and other PCMs to photoelectric detectors, leading to the realization that these materials could exhibit conductivity changes in response to light during phase transitions [36]. This phenomenon unlocked new opportunities for creating photodetectors that can regulate their performance based on illumination. With a deeper understanding of GST and other PCMs, researchers have effectively engineered prototype photodetectors utilizing these materials. These innovative detectors exhibit the capability to modulate and enhance current through controlled illumination, thereby demonstrating the promising application of PCMs in the field of optoelectronics [98].

One of the key advantages of PCMs is their application not only in photoelectric detectors but also in high-density, fast non-volatile memory, such as Phase Change Random Access Memory (PCRAM). These materials exploit the phase change properties to achieve non-volatile memory capabilities, significantly improving data storage efficiency [99, 100]. The non-volatile nature of PCMs eliminates the need for a continuous voltage supply, making them energy-efficient detectors [29]. Moreover, traditional

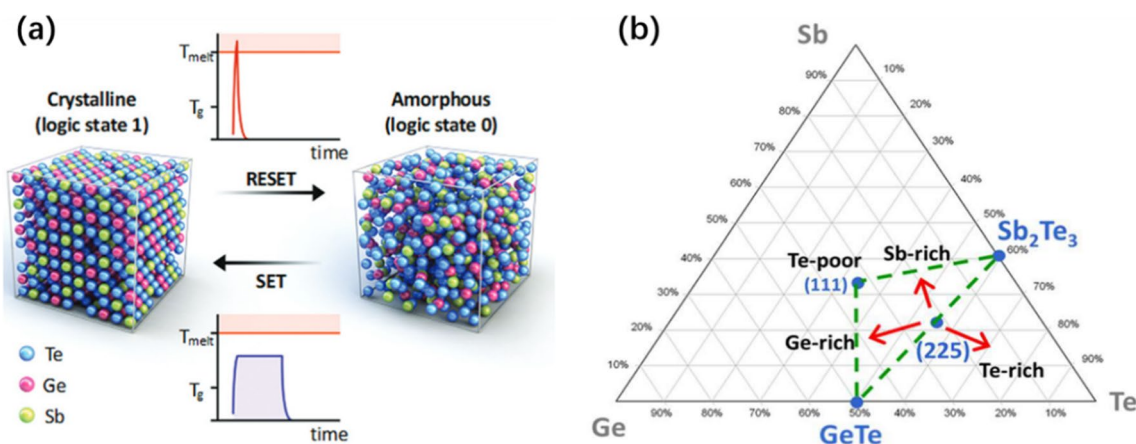


Figure 10 **a** Schematic diagram illustrating the working principle of PCMs, highlighting the reversible transition of the atomic structure between crystalline (low resistivity) and amorphous

(high resistivity) states [96]. **b** Ternary phase diagram of Ge–Sb–Te, optimized around GST material, demonstrates fast crystallization speeds and significant changes in optical reflectivity [97].

photodetectors often require separate light sources and broadband detection systems, a challenge that PCM-based photodetectors are better equipped to address by adjusting the phase of the material, these detectors can cover a more broad-spectrum range in the near-infrared spectrum, enhancing detection efficiency.

In recent years, PCMs such as GST, GeTe, and Sb_2S_3 have gained prominence in the field of photoelectric detection due to their advantages, including high sensitivity, rapid response times, controllability, low power consumption, long operational life, and a wide wavelength response range [101–103]. This part aims to showcase the superior performance of PCM-based photodetectors by comparing the photoelectric detection capabilities of three different PCM materials. The results presented in this part highlight the potential of PCMs to revolutionize the field of photodetection.

GST-based Photodetectors

Among the various PCMs discovered, $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST), a sulfur-based compound semiconductor PCM, has garnered more attention and widespread use in multiple applications, particularly in the field of photoelectric detection. GST is particularly well-suited for optoelectronic devices operating in the near-infrared range [104–106]. In its amorphous state, GST exhibits a band gap of approximately

0.65 eV, while its crystalline state has a smaller band gap of about 0.5 eV. This unique property facilitates tunable optical detection bandwidth and responsiveness, positioning GST as a versatile material for photodetectors.

To investigate the photoconductivity of nanoscale thin films of PCMs, Syed Ghazi Sarwat et al. [29] devised an optical cavity device comprising sputter-deposited thin films of ITO (indium tin oxide, 80 nm)/GST (21 nm)/ITO (40 nm)/Pt (50 nm), as illustrated in Fig. 11a. Their research also involved a comprehensive analysis of the behavior of these thin films. The findings revealed that the photoresponse of PCMs extends beyond photothermal effects, showcasing robust photoconductivity effects in both amorphous and crystalline states. Significantly, the study demonstrated the intricate interplay between photothermal and photoconductivity effects contributing to the overall photo response. The researchers highlighted the potential for fine-tuning these effects through strategic device engineering, involving optimizations in both area and electrode configurations. Photovoltaic effects were also documented within these devices, leading to deductions about the photocurrent mechanism. In the amorphous state, GST exhibited behavior akin to a p-type semiconductor due to Fermi pinning caused by localized defect states resulting from lone pairs, structural defects, and dangling bonds. In contrast, the crystalline state was identified as a degenerate p-type semiconductor, with the Fermi level placed within the valence band

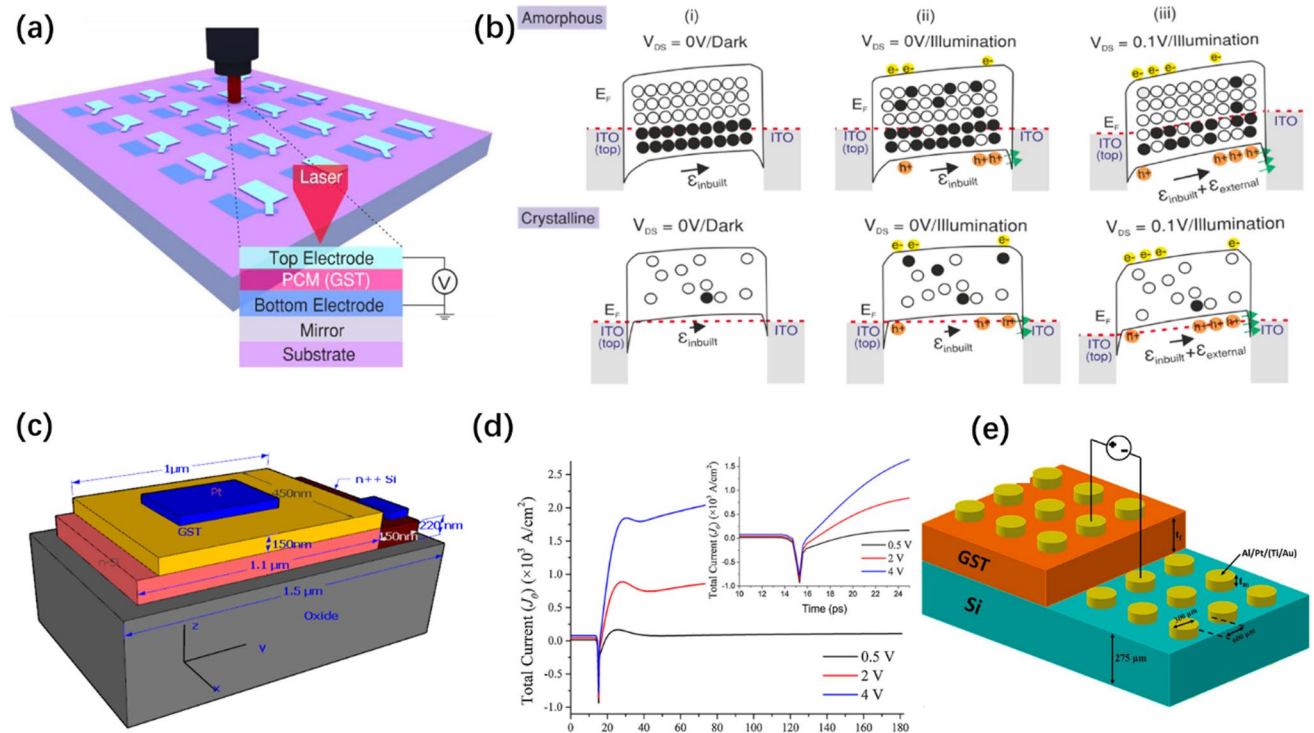


Figure 11 **a** Schematic illustration of the ITO/GST/ITO/Pt device [29]. **b** Photocurrent mechanism of A-GST and C-GST for the ITO/GST/ITO/Pt device [29]. **c** Schematic structure of an A-GST-based conventional vertical photodetector [37]. **d** Total current (J_0) for C-GST with a thickness of 150 nm at bias volt-

ages of 0.5 V, 2 V, and 4 V with a pulse rise time of 5 ps. The variations in J_0 during 10–25 ps are shown in the inset [30]. **e** Schematic diagram of a heterojunction with a GST and a silicon stack [31].

due to a very large hole concentration arising from structural vacancies, as illustrated in Fig. 11b. The noteworthy outcomes from these analyses include the tunability of state light detection bandwidth and light responsivity based on phase-change materials. This suggests a promising avenue for the development of tunable photodetectors, enhancing their adaptability and potential applications [107, 108].

Vibhu Srivastava et al. [37] first proposed a vertical photodetector based on amorphous $Ge_2Sb_2Te_5$ (A-GST), as depicted in Fig. 11c. This device, employed on silicon platforms at telecommunication wavelengths (1550 nm), exhibited a responsivity of $\sim 48 A W^{-1}$ and commendable light-to-dark current ratio following simulation and optimization. Subsequently, they proposed a vertical photodetector based on C-GST [109], optimized at an optimal thickness of ~ 210 nm, showcasing the highest responsivity of $\sim 63 A W^{-1}$ at 150 nm and a noteworthy responsivity of $\sim 41 A W^{-1}$ at 1550 nm, comparable to the performance of the A-GST-based device. The observed wavelength shift in the maximum responsivity further facilitated

the design of tunable photodetectors. Then, they investigated the impulse response of a photodetector based on GST integrated with a SOI waveguide using finite-difference time-domain simulations, as illustrated in Fig. 11d. Their findings demonstrated that a broad spectrum could be harnessed by tuning the device using the phase-change property of the material in the near-infrared region [30, 110]. Additionally, they developed a highly responsive phase-change GST visible light detector [111], noting that GST thin films prepared by RF sputtering exhibited responsiveness higher than $1 A W^{-1}$, responding to medium power visible spectra in both phases. Besides, the team creatively proposed a heterojunction with a GST and a silicon stack, as shown in Fig. 11e, enabling modulation in different operating spectra. The key factor in this switching was the utilization of the phase transition of GST, providing significant variations in optical absorption, resistivity, etc. The device proved useful in the NIR region, displaying photoconductivity modes and photothermal effects with good rectification at a low incident power of $3.38 \mu W$, a high responsivity of

$> 9 \text{ A W}^{-1}$, and a moderate detectivity of 2×10^9 Jones at a moderate bias of 3 V. The characteristics of GST deposited on various photodetector substrates were also evaluated by this team, establishing a distributed anti-reflective coating structure optimized for tunable photodetection performance based on PCMs [31, 112, 113], contributing significantly to the advancement of GST-based tunable photodetectors.

SPR is an important component of equipartition exciton photodetectors, crucial for enhancing the temperature sensitivity of the surface plasmon resonance sensors. In pursuit of improved sensitivity, Jiajun Xue et al. designed an ultra-high sensitivity fiber optic SPR temperature sensor based on a $\text{Ge}_2\text{Sb}_2\text{Te}_5$ -modified gold thin film [114, 115], as shown in Fig. 12a. Due to the high refractive index of the SPR sensor and the robust thermo-optical coefficient of cyclohexane, the SPR sensor demonstrates an average temperature sensitivity of up to $14.7 \text{ nm/}^\circ\text{C}$, a measurement range of

$5\text{--}80 \text{ }^\circ\text{C}$, and a temperature resolution of $1.36 \times 10^{-3} \text{ }^\circ\text{C}$. Notably, the highest temperature sensitivity reaches $25.7 \text{ nm/}^\circ\text{C}$, with a corresponding temperature resolution of $1.36 \times 10^{-3} \text{ }^\circ\text{C}$. This enhancement of SPR in sensitivity and resolution significantly improves the overall performance of equipartitioned exciton photodetectors, thereby enhancing the accuracy and reliability of detection.

Moreover, the integration of PCM-based photodetectors with 2D materials offers an avenue to further enhance sensitivity and response speed. The rapid responsiveness of PCMs to optical signals complements the high carrier mobility and surface area of 2D materials, synergistically improving the sensitivity and response speed of photodetectors. The electrical properties of PCMs can be regulated by changing their phase transition states, while 2D materials exhibit tunable photoelectric properties. The combination of these properties enables flexible adjustment of

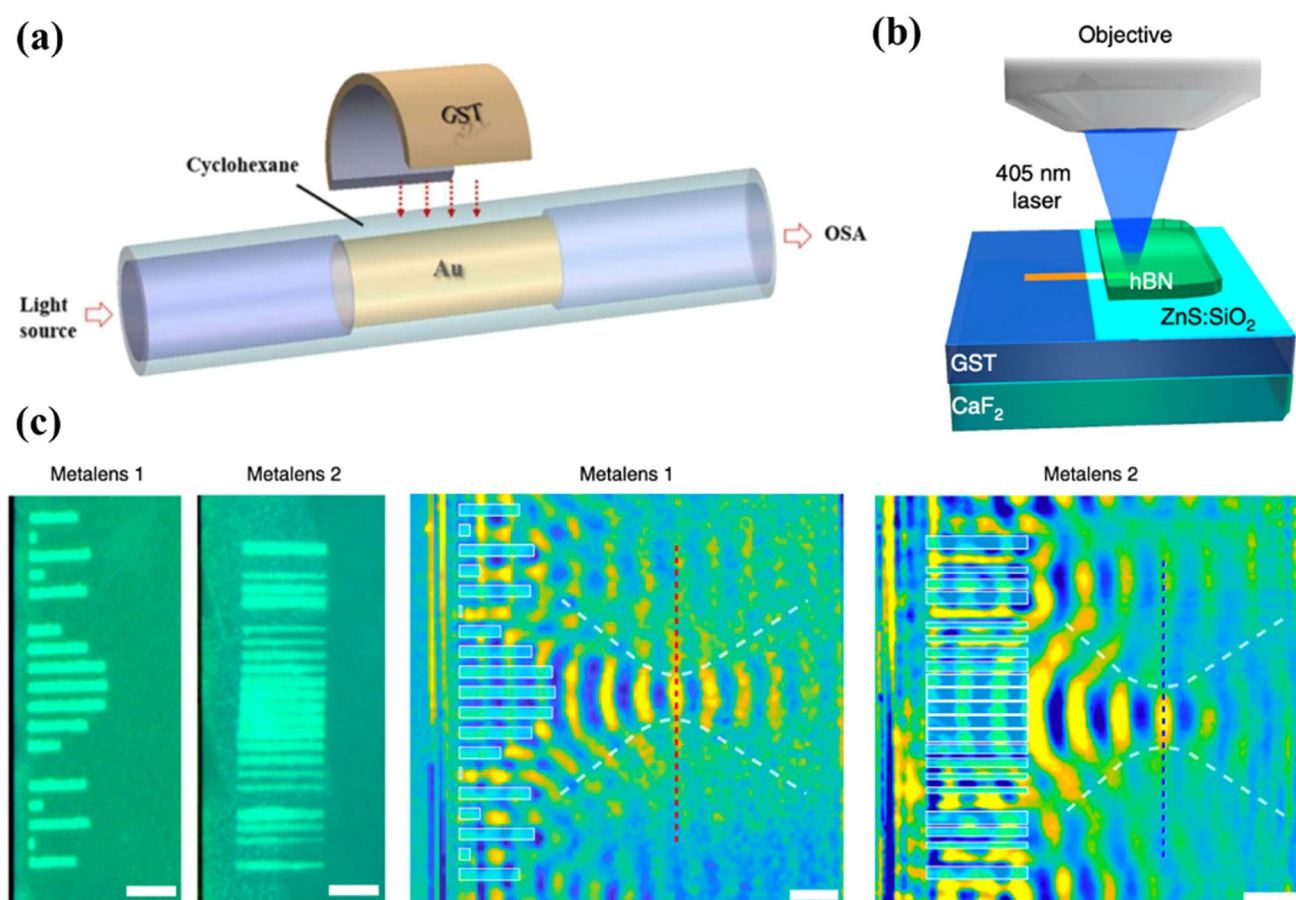


Figure 12 **a** Schematic illustration of a fiber optic SPR temperature sensor [114]. **b** Structural diagram of a hexagonal boron nitride device [116]. **c** Optical images of the fabricated metalenses, including s-SNOM images of metalens 2 at 1445 cm^{-1} [116].

photodetector performance. Kundan Chaudhary et al. [116] integrated hexagonal boron nitride (hBN) with GST, as illustrated in Fig. 12b. They demonstrate a metalens based on length changes, then they erase and replace it with a metalens where only the width of the individual elements varies. The scattering-type scanning near field optical microscopy (s-SNOM) characterization was carried out after each step and clearly reveals the focusing effect of the two designed lenses (Fig. 12c). This combination facilitates the fabrication of deep sub-wavelength polarized devices across the visible to infrared spectral range. The hBN-GST heterostructure serves as a versatile platform for precise control of polarization at the nanoscale, allowing for the creation of freeform morphology, transitions, and meta-optics. This approach enables cost-effective fabrication of fully-fledged polarized circuits, eliminating the need for conventional lithography. This low-cost implementation holds promise for biosensors and high-density optical storage, overcoming challenges posed by propagation losses to realize optical circuits and yielding practical devices.

Sb₂S₃-based photodetectors

In comparison to other PCMs, Sb₂S₃ stands out due to its notable attributes, including a large bandgap, high absorption coefficient, environmental friendliness, and abundance. Exhibiting good photosensitivity with a bandgap spanning the entire visible spectrum (1.5–2.2 eV), antimony sulfide is regarded as a promising optoelectronic material [117–120].

Gonzalo Santos et al. [32] delved into the exploration of surface plasmons in metal plasma nanostructures, culminating in the design of a reconfigurable photodetector that integrates Au grating with the phase-change semiconductor Sb₂S₃, as illustrated in Fig. 13a. The proposed structure of the reconfigurable photodetector exhibits distinctive characteristics, with the wavelength refractive index characteristic curve presented in Fig. 13b. The modulation of surface plasmons (SPs) characteristics at the double-comb Au electrode is contingent upon the refractive index variation between the amorphous and crystalline phases of Sb₂S₃. Notably, a clear refractive index difference in photoelectric detection is evident in the Sb₂S₃ curve, as observed in Fig. 13b, particularly from the band at $\lambda = 1310$ nm to the band at $\lambda = 1550$ nm. In plasma nanostructures, the efficiency of hot electrons can be optimized by maximizing light absorption, thus

optimizing the production of hot electrons. They also conducted detailed numerical studies of the Au grating parameters and SP order formed by the double-comb electrodes. This comprehensive investigation aims to enhance the absorption capabilities of the device, consequently optimizing the responsivity of the photodetector.

Due to the enhanced stability of Au and Sb₂S₃ compared to other metals, a photodetector with an Au-Sb₂S₃-Au structure was designed [121], and its configuration is depicted in Fig. 13c. Sb₂S₃ undergoes a reversible transition from a crystalline state to an amorphous state through photothermal or electrothermal heating at temperatures exceeding 573 K (crystalline phase) and 801 K (amorphous phase). This transformation, from the crystalline phase to the amorphous phase, is characterized by intriguing attributes, including an ultra-fast switching speed of approximately 70 ns and a substantial bandgap. The substantial bandgap of Sb₂S₃ makes it an ideal candidate for tunable photonics applications in many visible spectral ranges, as it shows relatively low losses, and both states are stable at room temperature.

Kai Zhao et al. systematically investigated the optical, vibration and photoelectron anisotropy of high quality orthogonal Sb₂S₃ nanowires through experimental and theoretical studies [122]. High quality Sb₂S₃ nanowires were synthesized by sulfur assisted gas transport. An atomic force microscope image shows a typical Sb₂S₃ nanowire. The crystal structure of Sb₂S₃ is shown in Fig. 13d. The metal-semiconductor-metal photodetector based on a single Sb₂S₃ nanowire has good polarization sensitivity in a wide band range from ultraviolet to near-infrared (360–1550 nm), with a peak dichroic ratio of 2.54 at 638 nm.

Kun Ye et al. [123] comprehensively and systematically characterized the light detection performance of Sb₂S₃ nanowires. The Sb₂S₃ photodetector exhibits a wide band optical response from ultraviolet (360 nm) to near-infrared (785 nm), with temporal optical responses at different wavelengths portrayed in Fig. 13e. The high quality Sb₂S₃ nanowires prepared by CVD method have great application potential in semiconductor and optoelectronics. In order to make the photodetector based on Sb₂S₃ have a wider detection range, Ju Hung Chen et al. [124] introduced graphene, a 2D material capable of adjusting the response wavelength range based on its band gap size. The schematic diagram in Fig. 13f illustrates the Sb₂S₃ film deposited

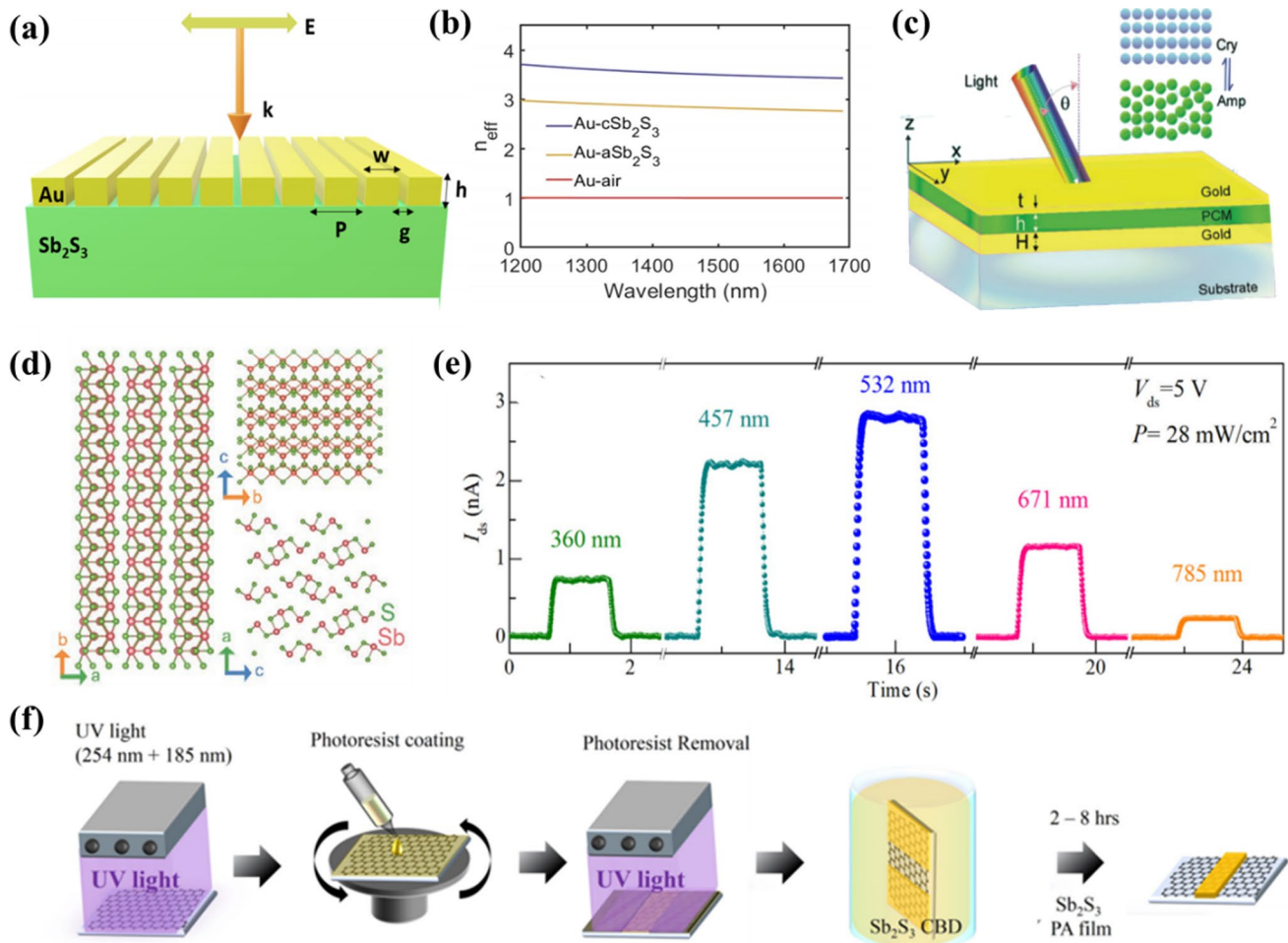


Figure 13 **a** Diagram illustrating a reconfigurable metal–semiconductor configuration with interdigitated dual-comb gold electrodes in the photodetector structure [32]. **b** Depiction of refractive index changes in the reconfigurable photodetector at various wavelengths [32]. **c** Crystal structure of Sb_2S_3 nanowires [121].

d Characterization of the Sb_2S_3 crystal structure observed from three coordinate axes [122]. **e** Time-dependent light response of Sb_2S_3 photodetector at different wavelengths [123]. **f** Schematic diagram of the Sb_2S_3 thin film deposited on functionalized graphene [124].

on functionalized graphene after photolithography. By taking advantage of the tunable properties of graphene, efficient chemical deposition is achieved, and defects induced by ozone in graphene are instrumental in improving material morphology and quality. The optimal deposition process is designed by comprehensive characterization of the seed layer and Sb_2S_3 films. This material demonstrates unprecedented responsiveness and detection capabilities in semiconductor/graphene/glass vertical heterostructures, featuring efficient carrier transport and high photodetector performance (112 A W^{-1} , 2.01×10^{12} Jones).

GeTe-based photodetectors

Currently, GeTe has not seen extensive development in the realm of photoelectric detection. However, it exhibits characteristics conducive to high-performance photoelectric detection under specific light intensities, suggesting significant potential in the field [125]. The resistivity in the crystalline state of GeTe remains unaffected by annealing temperature, with resistivity variations occurring within a narrow temperature range. Figure 14a illustrates the resistivity variations of GeTe at different temperatures. GeTe is generally recognized as a degenerate semiconductor, wherein the Fermi energy resides within the valence band, imparting metal-like behavior

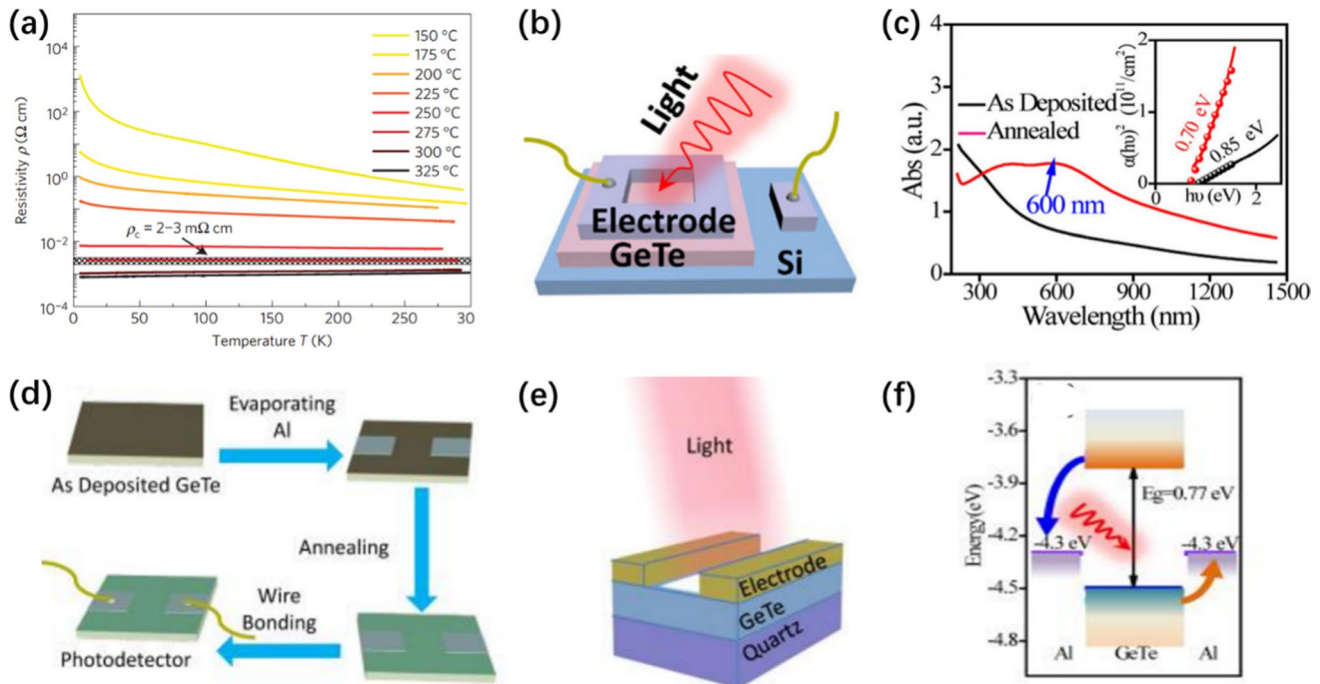


Figure 14 **a** Resistivity of GeSb₂Te₄ film between 5 and 300 K [126]. **b** Structure of the p-GeTe/n-Si heterojunction photodetector [127]. **c** Ultraviolet–visible–near-infrared absorption spectra of GeTe films before and after annealing [127]. **d** Schematic

diagram of photodetector manufacturing process based on GeTe thin film. **e** Schematic diagram of the GeTe photodetector device [128]. **f** Energy band diagram illustrating the working principle of the photodetector [128].

and consequently a metallic temperature coefficient of resistivity. In GeSb₂Te₄, crystallization is also marked by a sharp decrease in resistivity. However, unlike GeTe, the resistivity in the crystalline state of GeSb₂Te₄ exhibits a clear dependence on annealing [126].

The absorption coefficient of GeTe measures approximately 10^5 cm^{-1} . When coupled with the high carrier exhibited by GeTe thin films, this feature holds promise for the fabrication of infrared photodetectors. Yiqun Zhao et al. simulated a prototype photovoltaic detector based on a p-GeTe/n-Si heterojunction [127, 128]. The structure of the photodetector, as shown in Fig. 14b, indicates the potential application of GeTe in infrared detection. GeTe presents substantial opportunities for integration into other fields, particularly computing and storage, leveraging GeTe-based materials. In the experimental setup, the N-type monocrystalline silicon substrate underwent cleaning using a chemical bath method and subsequent air precipitation. Subsequently, a GeTe film was directly deposited on the pristine substrate via magnetron sputtering under a specific pressure vacuum condition. The

film was then enveloped in copper foil and annealed in a vacuum oven at 360 °C for 10 min. UV–visible–near-infrared absorption spectra of GeTe films before and after annealing are presented in Fig. 14c. Finally, a pair of Al electrodes were evaporated onto the GeTe film and Si substrate through a shadow mask using physical vapor deposition technique (at a pressure of $7.0 \times 10^{-5} \text{ Pa}$). The fabrication process diagram of the photodetector based on GeTe thin film is shown in Fig. 14d. In Fig. 14e, the structure of the photodetector is presented, consisting of GeTe and Al layers with thicknesses of 35 nm and 100 nm, respectively. In this study, the values of E_g for the GeTe films were determined to be 0.85 eV before annealing and 0.77 eV after annealing. The operational principle of the photodetector is clarified through the energy band diagram presented in Fig. 14f. The process can be delineated into two sequential steps: the absorption of photons by the GeTe film and the subsequent transfer of charge carriers to the Al electrodes. The incorporation of GeTe’s distinctive storage and computing features is poised to exert a substantial influence in the realm of computing. Given its status as a narrow band gap semiconductor, GeTe materials hold promise

for application in infrared detectors. A breakthrough in utilizing GeTe in photodetectors could establish a pivotal connection between the fields of photoelectric detection and computing in the future.

Conclusion and prospect

In conclusion, this review provides a thorough examination of recent advancements and future prospects in photodetectors. It discusses traditional inorganic semiconductor materials such as Si, GaN, and InGaAs, alongside emerging 2D materials like graphene, perovskite and MXene. The review also explores photodetectors enhanced by SPR technology, utilizing oxides, nitrides, and 2D materials. Integration of SPR significantly enhances photodetector performance, leading to promising developments in optoelectronics. Additionally, the review investigates photodetectors utilizing PCMs like GST, GeTe, and Sb_2S_3 , known for their tunable wavelengths. The combination of plasmonic enhancements with PCMs results in photodetectors offering adaptable detection rates and minimal losses. This synergy leverages the tunable wavelengths of PCMs to precisely control detection, opening new possibilities in photodetector design and functionality.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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